

Solution Tutorial III – Stability

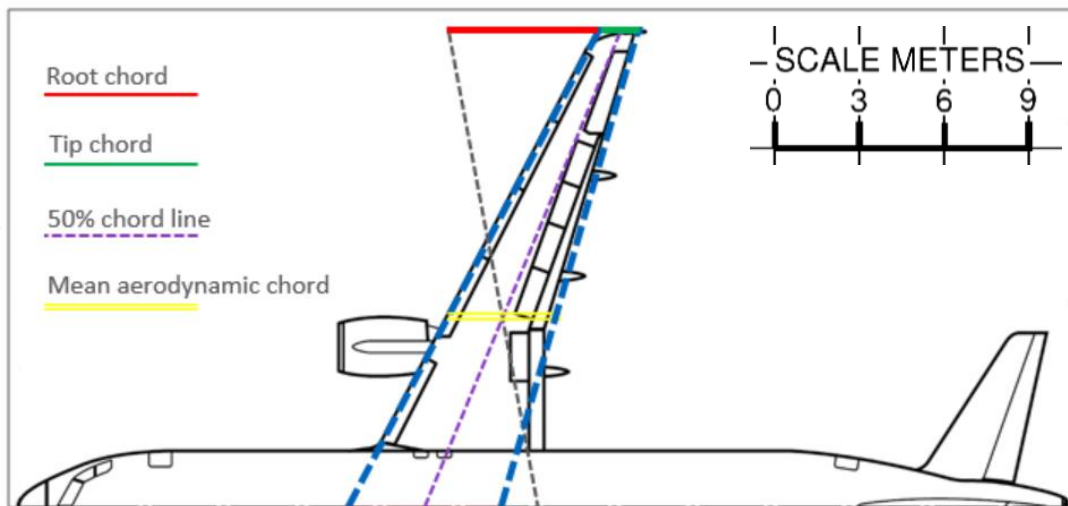
Task 1

Determine the mean aerodynamic chord c_{MAC} of the Airbus A320. Apply the algebraic method and the geometric method (approximate the wing with a trapezoidal shape to be able to use these methods). Compare the results with the manufacturer's value.

- Algebraic method:

$$c_{MAC} = \frac{2}{3} \cdot c_{root,W} \cdot \frac{1 + \lambda + \lambda^2}{1 + \lambda} = \frac{2}{3} \cdot 6.07 \cdot \frac{1 + 0.2471 + 0.2471^2}{1 + 0.2471} = 4.245m$$

- Geometric method:



Read from figure: $c_{MAC} \cong 4.2m$

- Manufacturer: $c_{MAC} = 4.1935 m$

Please be aware that both presented methods only work for trapezoidal wings without a kink. Hence, above we approximated the wing with a trapezoid. For other wing geometries the following equation should be used: $c_{MAC} = \frac{2}{S_{ref}} \int_0^{b/2} c(y)^2 dy$

The following tasks (2-5) are carried out with the goal to eventually calculate the aircraft's neutral point. Therefore, the individual terms of the following equation will be calculated:

$$\bar{x}_{AC} = \frac{C_{L\alpha,W} \cdot \bar{x}_{AC,W} + C'_{L\alpha,HT} \cdot \frac{q_{HT}}{q} \cdot \frac{S_{HT}}{S_{ref}} \cdot \frac{\partial \alpha_{HT}}{\partial \alpha} \cdot \bar{x}_{AC,HT} - C_{M\alpha,fus}}{C_{L\alpha,W} + C'_{L\alpha,HT} \cdot \frac{q_{HT}}{q} \cdot \frac{S_{HT}}{S_{ref}} \cdot \frac{\partial \alpha_{HT}}{\partial \alpha}}$$

Task 2

Calculate the lift coefficient slope $C_{L\alpha,j}$ of each aircraft assembly (wing, horizontal tail and fuselage) at $M = 0.78$. Scale and correct $C_{L\alpha,HT}$ to aircraft level.

$$\beta = \sqrt{1 - M^2} = \sqrt{1 - 0.78^2} = \mathbf{0.6258}$$

- Main wing (see also tutorial 1, task 3):

$$K_W = 1 + 0.75 \cdot \left(\frac{t}{c}\right)_W = 1 + 0.75 \cdot 11.75\% = \mathbf{1.088}$$

$$\begin{aligned} \tan(\Lambda_{50\%,W}) &= \tan(\Lambda_{25\%,W}) - \frac{4}{AR_W} \cdot \left[\frac{50 - 25}{100} \cdot \frac{1 - \lambda_W}{1 + \lambda_W} \right] \\ \Rightarrow \tan(\Lambda_{50\%,W}) &= \tan(0.4356 \text{ rad}) - \frac{4}{9.39} \cdot \left[\frac{50 - 25}{100} \cdot \frac{1 - 0.2471}{1 + 0.2471} \right] = \mathbf{0.4011} \end{aligned}$$

$$\begin{aligned} C_{L\alpha,W} &= \frac{2\pi \cdot AR_W}{2 + \sqrt{\frac{AR_W^2 \cdot \beta^2}{K_W^2} \cdot \left(1 + \frac{\tan^2 \Lambda_{50\%,W}}{\beta^2}\right)} + 4} \\ \Rightarrow C_{L\alpha,W} &= \frac{2\pi \cdot 9.39}{2 + \sqrt{\frac{9.39^2 \cdot 0.6258^2}{1.088^2} \cdot \left(1 + \frac{0.4011^2}{0.6258^2}\right)} + 4} = \mathbf{6.766 /rad} \end{aligned}$$

- Horizontal tail:

$$K_{HT} = 1 + 0.75 \cdot \left(\frac{t}{c}\right)_{HT} = 1 + 0.75 \cdot 10\% = \mathbf{1.075}$$

$$\begin{aligned} \tan(\Lambda_{50\%,HT}) &= \tan(\Lambda_{25\%,HT}) - \frac{4}{AR_{HT}} \cdot \left[\frac{50 - 25}{100} \cdot \frac{1 - \lambda_{HT}}{1 + \lambda_{HT}} \right] \\ \Rightarrow \tan(\Lambda_{50\%,HT}) &= \tan(0.5061 \text{ rad}) - \frac{4}{5} \cdot \left[\frac{50 - 25}{100} \cdot \frac{1 - 0.2560}{1 + 0.2560} \right] = \mathbf{0.4358} \end{aligned}$$

$$\begin{aligned} C'_{L\alpha,HT} &= \frac{2\pi \cdot AR_{HT}}{2 + \sqrt{\frac{AR_{HT}^2 \cdot \beta^2}{K_{HT}^2} \cdot \left(1 + \frac{\tan^2 \Lambda_{50\%,HT}}{\beta^2}\right)} + 4} \\ \Rightarrow C'_{L\alpha,HT} &= \frac{2\pi \cdot 5}{2 + \sqrt{\frac{5^2 \cdot 0.6258^2}{1.075^2} \cdot \left(1 + \frac{0.4358^2}{0.6258^2}\right)} + 4} = \mathbf{5.174 /rad} \end{aligned}$$

$$C_{L\alpha,HT} = C'_{L\alpha,HT} \cdot \frac{S_{HT}}{S_{ref}} \cdot \left(q_{HT}/q \cdot \frac{\partial \alpha_{HT}}{\partial \alpha} \right)$$

Scaling

Correction

$$\Rightarrow C_{L\alpha,HT} = 5.17 \cdot \frac{31}{122.4} \cdot (0.85 \cdot 0.65) = 0.7240 /rad$$

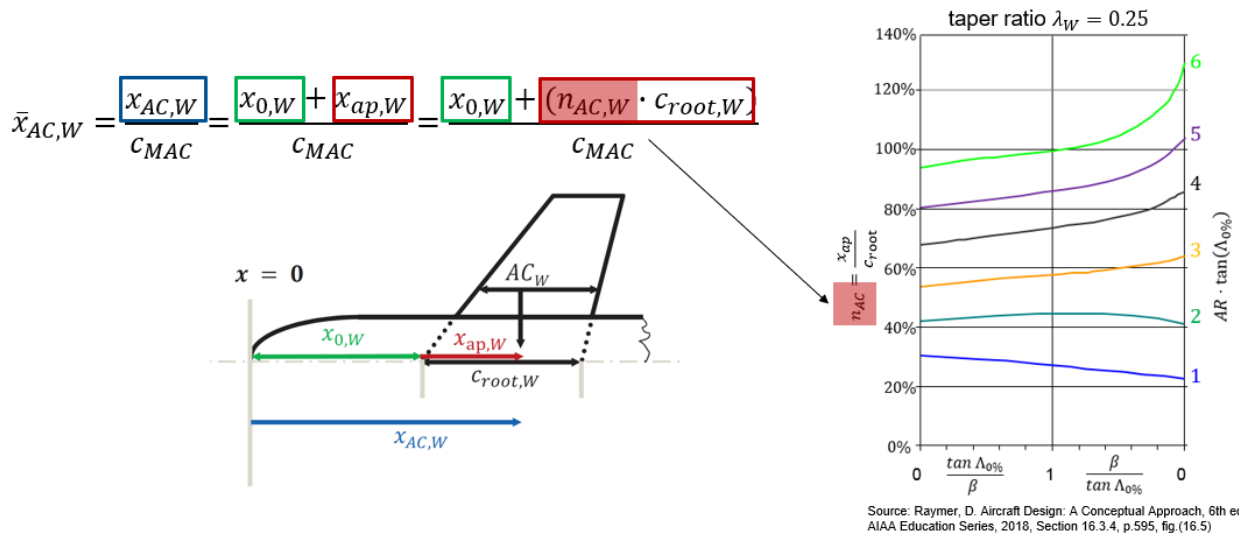
- Fuselage:

$$C_{L\alpha,fus} = 0 /rad \text{ (neglect in first approximation)}$$

Task 3

Determine the normalized aerodynamic center of the main wing $\bar{x}_{AC,W}$ and the horizontal tail $\bar{x}_{AC,HT}$. The fixed distance from the nose of the aircraft to each apex can be found in the aircraft dataset.

General approach:



How to read n_{AC} from the diagram:

1. Choose the right diagram acc. to the taper ratio
2. Calculate $AR \tan \Lambda_{0\%,W}$ and round the result to an integer
3. Choose the respective curve
4. Calculate $\frac{\tan \Lambda_{0\%,W}}{\beta}$ and $\frac{\beta}{\tan \Lambda_{0\%,W}}$
5. Choose side for which the value is between 0 and 1
6. Read n_{AC} from diagram

- Wing:

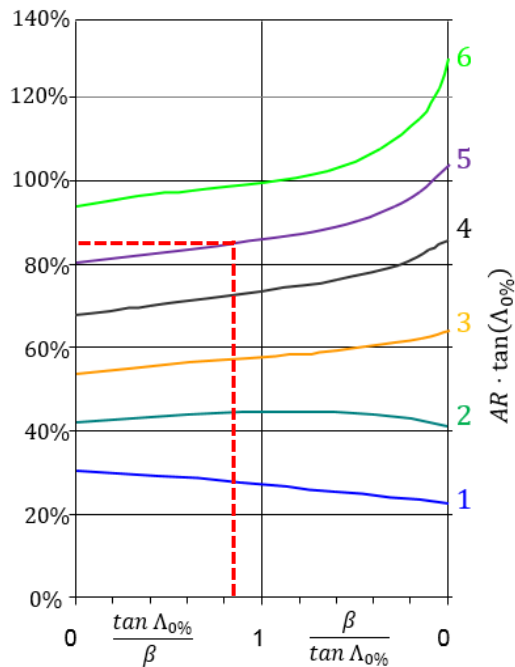
Taper ratio $\lambda_W = 0.2471 \approx 0.25 \rightarrow$ choose figure accordingly

$$\tan(\Lambda_{0\%,W}) = \tan(\Lambda_{25\%,W}) - \frac{4}{AR_W} \cdot \left[\frac{0 - 25}{100} \cdot \frac{1 - \lambda_W}{1 + \lambda_W} \right]$$

$$\Rightarrow \tan(\Lambda_{0\%,W}) = \tan(0.4356 \text{ rad}) - \frac{4}{9.39} \cdot \left[\frac{0 - 25}{100} \cdot \frac{1 - 0.2471}{1 + 0.2471} \right] = 0.5297$$

curve: $AR_W \cdot \tan(\Lambda_{0\%,W}) = 9.39 \cdot 0.5297 = 4.98 \approx 5$

→ Purple line



x-axis:

$$\frac{\tan \Lambda_{0\%,W}}{\beta} = 0.8464$$

$$\frac{\beta}{\tan \Lambda_{0\%,W}} = 1.181$$

→ Take value that's between 0 and 1

y-axis:

$$n_{AC,W} \cong 85.9\%$$

$$\bar{x}_{AC,W} = \frac{x_{0,W} + (n_{AC,W} \cdot c_{root,W})}{c_{MAC}}$$

$$\Rightarrow \bar{x}_{AC,W} = \frac{11.76 + (85.9\% \cdot 6.07)}{4.1935}$$

$$= 4.048$$

- Horizontal tail:

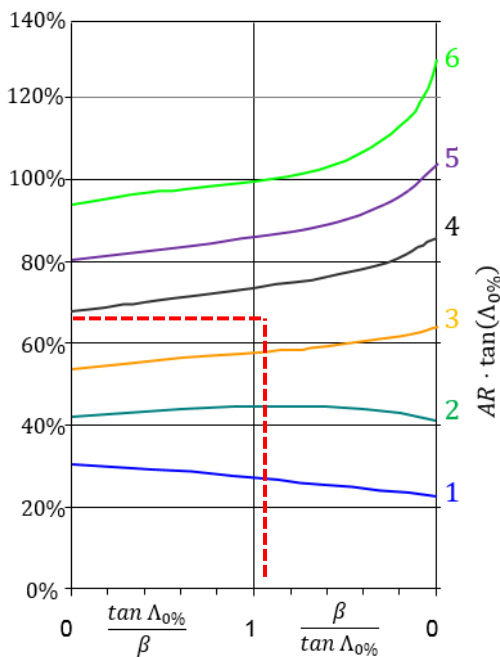
Taper ratio $\lambda_{HT} = 0.2560 \approx 0.25 \rightarrow$ choose figure accordingly

$$\tan(\Lambda_{0\%,HT}) = \tan(\Lambda_{25\%,HT}) - \frac{4}{AR_{HT}} \cdot \left[\frac{0 - 25}{100} \cdot \frac{1 - \lambda_{HT}}{1 + \lambda_{HT}} \right]$$

$$\Rightarrow \tan(\Lambda_{0\%,HT}) = \tan(0.5061 \text{ rad}) - \frac{4}{5} \cdot \left[\frac{0 - 25}{100} \cdot \frac{1 - 0.2560}{1 + 0.2560} \right] = 0.6727$$

curve:

$$AR_{HT} \cdot \tan(\Lambda_{0\%,HT}) = 5 \cdot 0.6727 = 3.363 \approx 3$$



x-axis:

$$\frac{\tan \Lambda_{0\%,HT}}{\beta} = 1.075$$

$$\frac{\beta}{\tan \Lambda_{0\%,HT}} = 0.9303$$

$\rightarrow$ Take value that's between 0 and 1

y-axis:

$$n_{AC,HT} \cong 58.4\%$$

$$\bar{x}_{AC,HT} = \frac{x_{0,HT} + (n_{AC,HT} \cdot c_{root,HT})}{c_{MAC}}$$

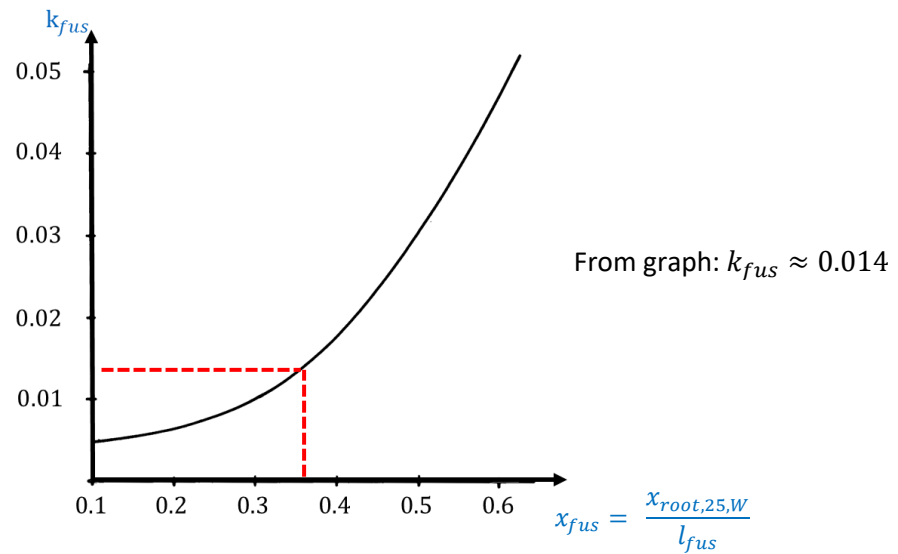
$$\Rightarrow \bar{x}_{AC,HT} = \frac{31.55 + (58.4\% \cdot 3.74)}{4.1935} = 8.044$$

Task 4

Determine the moment coefficient slope $C_{M\alpha, fus}$ of the fuselage.

- Fuselage:

$$x_{fus} = \frac{x_{root,25,W}}{l_{fus}} = \frac{x_{0,W} + 0.25 \cdot c_{root,W}}{l_{fus}} = \frac{11.76 + 0.25 \cdot 6.07}{37.57} = 0.3534$$

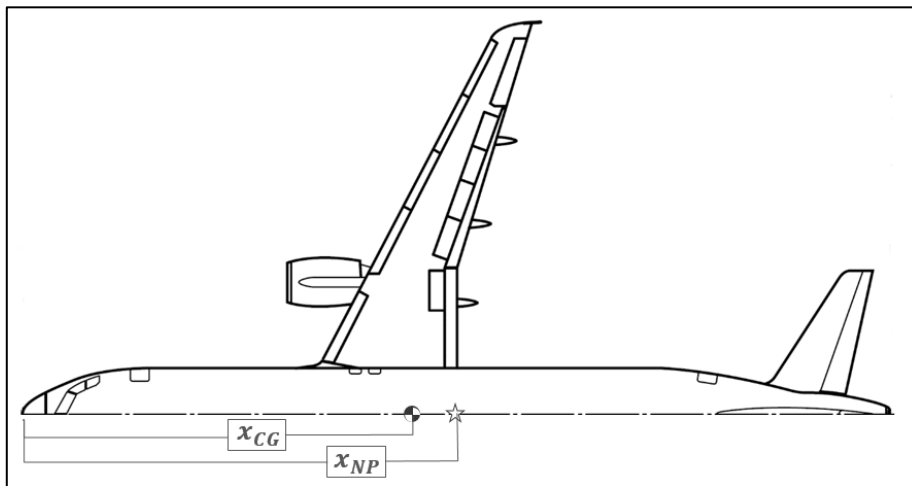


$$C_{M\alpha,fus} = 45 \cdot \frac{k_{fus} \cdot w_{fus}^2 \cdot l_{fus}}{c_{MAC} \cdot S_{ref}} = 45 \cdot \frac{0.014 \cdot 3.95^2 \cdot 37.57}{4.1935 \cdot 122.4} = 0.7195/rad$$

Task 5

Finally, locate the aircraft's aerodynamic center $\bar{x}_{AC}$ and compute the stability measure, assuming that the center of gravity location x_{CG} in cruise is at 90% of root chord.

Aircraft aerodynamic center $\bar{x}_{AC}$ = aircraft neutral point $\bar{x}_{NP}$



$$\bar{x}_{AC} = \frac{C_{L\alpha,W} \cdot \bar{x}_{AC,W} + C_{L\alpha,HT} \cdot \bar{x}_{AC,HT} - C_{M\alpha,fus}}{C_{L\alpha,W} + C_{L\alpha,HT}} = \frac{6.766 \cdot 4.048 + 0.7240 \cdot 8.044 - 0.7195}{6.766 + 0.7240} = 4.338$$

$$\bar{x}_{CG} = \frac{x_{CG}}{c_{MAC}} = \frac{x_{0,W} + 90\% \cdot c_{root,W}}{c_{MAC}} = \frac{11.76 + 0.9 \cdot 6.07}{4.1935} = 4.107$$

$$\frac{\partial C_M}{\partial C_L} = \bar{x}_{CG} - \bar{x}_{AC} = 4.107 - 4.338 = -\mathbf{23.10\%}$$

Task 6

Determine the horizontal tail volume V_{HT} of the aircraft. Compare your result with the horizontal tail volume of similar aircraft.

$$\begin{aligned} V_{HT} &= \frac{S_{HT}}{S_{ref}} \cdot \frac{d_{HT}}{c_{MAC}} = \frac{S_{HT}}{S_{ref}} \cdot \frac{x_{AC,HT} - x_{AC,W}}{c_{MAC}} = \frac{S_{HT}}{S_{ref}} \cdot \bar{x}_{AC,HT} - \bar{x}_{AC,W} \\ &\Rightarrow V_{HT} = \frac{31}{122.4} \cdot (8.044 - 4.048) = \mathbf{1.012} \end{aligned}$$

Compare with similar aircraft:

$$\frac{S_{HT}}{S_{ref}} \approx 0.2 - 0.35$$

$$V_{HT} = \frac{S_{HT}}{S_{ref}} \cdot \frac{d_{HT}}{c_{MAC}} \approx 0.5 - 1.0$$